

Defense Brief — Representation Convergence Project

Companion to the 74-page report. One page of framing per topic; every number cited is measured and appears in the report with its generator notebook.

1 · Assumptions (what the results are conditional on)

Item-level correspondence exists and is known for supervised fits. Row i of every array is the same passage or image in every model. This is constructed (nested sorted-id prefixes, deterministic caption recipe) and verified numerically wherever two caches meet; C.12 is the one experiment that deliberately removes it.

Linear maps, closed form. Every fitted map is ridge regression solved in closed form - no gradients, no iteration - so 'related by a linear transformation' carries no asterisk. The MLP contrast (F2) is the deliberate exception and converged to the linear answer exactly.

Frozen encoders. No pretrained weight is modified anywhere in the evidence chain; F3, which would unfreeze one, is built but gated shut to preserve that statement.

Held-out discipline includes the transforms. Splits precede fitting; whitening covariances, hub bases, and alpha choices come from training rows only. Gallery size is 1,000 everywhere, so chance is 0.001 and R@K columns are comparable across experiments.

Geometry diagnostics are trusted as measurements. Effective rank (spectral entropy) and mean pair-cosine are treated as valid summaries of a space's shape; C.14's perfect rank ordering is itself evidence the trust is earned.

2 · Restrictions (scope the claims do not exceed)

claim	measured scope	explicitly NOT claimed
convergence is linear	GPT-2/Pythia; MobileCLIP/SigLIP; DINOv2 x3; 4 text encoders	all architectures, all scales
alignment scales with capacity	DINOv2 family, 22M-304M (13.8x), compressive at the top	behaviour beyond 304M; other families
isotropy rescue	GPT-2 target, two image scales (102.6%/101.4%)	sufficiency - C.14 bounds it
unsupervised recovery	39.7% exact @ 595x chance; 40% of ceiling	a deployable pairing-free system
shared hub	5 encoders, 2 families, COCO, English; transfer 93-97%, cross-lineage and cross-architecture	a universal space for all models
deployment numbers	COCO val/train2017 domain; 1,000-item galleries	open-world scale or other domains

One language, one image domain, one caption corpus throughout. Every 'percent of ceiling' is against a ceiling re-measured under the same protocol and caption treatment - never a borrowed number.

3 · Tuning (every knob, and how it was set)

knob	how set	guard against overfitting the knob
ridge alpha	validation sweep 1e-3..10 per experiment	chosen on train-internal validation, never on eval
pooling	fixed cls+patch across the scale sweep	the one mixed run (base CLS) excluded from comparisons
N (pairs)	scaled with encoder width -> 11.1 rows/dim at every sweep point	starvation biases DOWN, so it cannot fake the trend
GW epsilon	swept per run; optimum MOVES with N (5e-3@800 -> 1e-3@1500)	reported as a sweep, never a single tuned value
HUB_DIM x alpha	2-D sweep, transfer-first, rows/dim ≥ 5 enforced	pre-registered thresholds; cliff reported as finding
gates	90%-of-ceiling; 85% transitivity; 50% transfer; ≥ 5 rows/dim	registered before results; 89.0% reported as NOT met

4 · Defense — the hard questions, with answers

“Isn't whitening cheating?” It is label-free (uses no pairs, no labels), fitted on training targets only, and linear - so it is an instance of the thesis, not an exception. The full map is whitened composed with ridge: still one matrix.

“Why is the whitened R-squared 0.014?” Whitening destroys magnitudes and perfects angles. R-squared asks about magnitudes; retrieval asks about angles. One system, two metrics, opposite verdicts - which is the project's metric-geometry lesson in its purest form.

“You said isotropy helps - then why does it destroy C.12?” Reading versus finding. Cosine reads a KNOWN correspondence: uneven variance is noise. Gromov-Wasserstein FINDS an unknown one from distance structure: uneven variance is the signal that makes points distinctive. Same property, opposite prescriptions - both measured, twice each.

“Doesn't C.14's residual spread undermine C.11?” It bounds it. C.11's mechanism predicted two unseen encoders' ordering perfectly (rank-corr +1.00). The residue after whitening (0.110) measures what contrastive training adds BEYOND the frame: the arrangement of meaning across directions. Necessary, not sufficient - a stronger theory than the original.

“Why did the hub break between 512 and 768 dimensions?” Least squares divided by a near-null singular value. The head's coordinates have algebraic rank 768, so at hub width 768 the LAST direction is the vanishing one - sigma 0.0054 against 10.79 one step earlier - and the fit divides by it. Seven accounts were falsified first; the eighth predicted the location on two out-of-sample encoders and then resolved to a single direction: transfer holds to k=760 and collapses at 768. Truncation or a larger head alpha repairs it. So the cliff is a numerical artifact of the fit, not a property of shared coordinates - and 512 is safe because it is below the source's rank.

“Is the transfer result circular - the head just learned the task?” The random-map control replaces only the hub map and collapses to chance (0.001), so the head alone carries nothing. And absolute R@1 RISES on unseen encoders, which a task-memorising head cannot produce.

“Have the artifacts themselves been checked?” Yes, in one automated run after all experiments were complete: 47 checks across every stored artifact - statistical power, row alignment by shuffle test, sanity, target geometry, and cache-key integrity. NO hard failures. Every fit sits between 5.9 and 9.8 rows per input dimension; shuffle gaps run 0.545 to 0.863 against a 0.2 threshold; all three encoder caches match the configuration named in their filenames. The only two flags are the GPT-2 caption space, whose collapse is a documented finding rather than a defect - and the battery reproduced its signature independently in two separately generated files. (Appendix D.5.)

“How do I know the numbers are right?” Five documented self-corrections via measurement (starvation rerun, synthetic-figure retraction, matched ceiling, wrong reference arm twice - now derived from filenames, whitening-for-GW reversal), plus width-tagged checkpoints, transfer checks against known pairs, and AST-validated notebooks. The error CLASS is real; the defenses are structural.

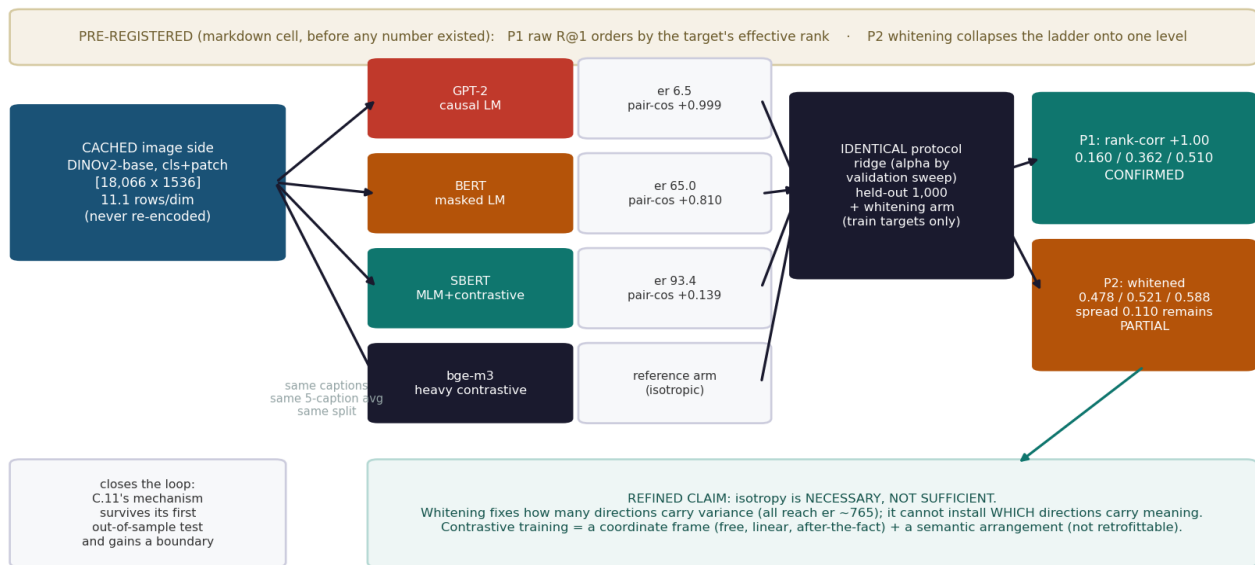
5 · Achievements — the nine findings

#	finding	headline evidence	section
1	convergence is linear	MLP = ridge exactly at 74 rows/dim (0.9052 = 0.9052)	C.9/F
2	related, not rigid	Procrustes 0.038 vs ridge 0.592; 69x singular-value spread	4, B
3	deployable	95% of ceiling @ 0.79 MB; verif AUC 0.999 / rank 0.987 / membership 0.889	4, D, D.2
4	cross-modal, no joint training	40.9% of matched ceiling, 358x chance	C.8-C.9
5	alignment scales with capacity	+15.3 pts of ceiling per decade, 6 monotone measures, compressive	C.10
6	isotropy, not similarity training	whitening -> 102.6% / 101.4% of bge at two scales; rho +0.94	C.11
7	correspondence is self-identifying	GW 39.7% exact = 595x chance, 0 pairs; + 2 measured negatives	C.12
8	one shared space, used	hub transfer 95.9%/92.9% of native on unseen encoders; control at chance	C.13
9	the mechanism predicts	pre-registered ladder: rank-corr +1.00; isotropy necessary-not-sufficient	C.14

Plus the recurring meta-finding, measured **TWICE** : isotropy helps reading a correspondence and hurts finding or losslessly carrying one (C.11 vs C.12 vs the C.13 width cliff).

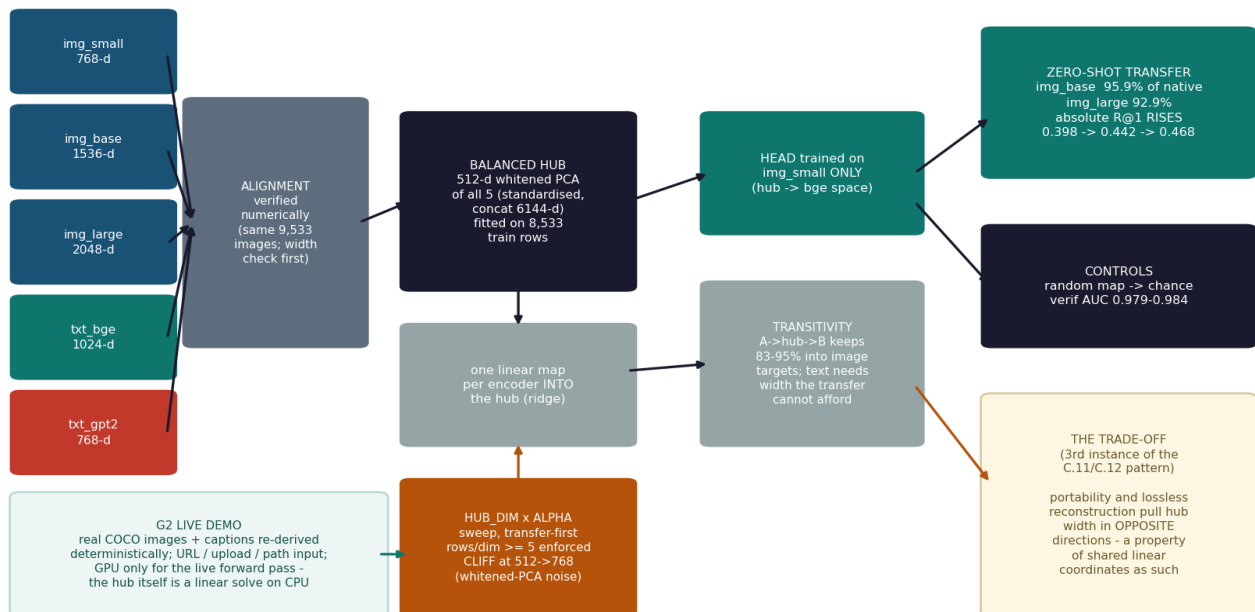
6 · Flow: how E1.3 closes the loop

E1.3 — the encoder ladder: one image side, four text objectives, predictions registered first



Flow: the hub, built and demonstrated

G1/G2 — five encoders, one 512-d hub: the shared space built, measured, and demonstrated live



7 · Conclusion

The thesis - independently trained models share representations up to a linear transformation - was tested in every posture available: pairwise and supervised (it holds, and an MLP adds exactly nothing); across modalities without joint training (it holds at 41-58 of a jointly-trained ceiling, rising with capacity at a measured slope); with the metric geometry itself interrogated (apparent failures were coordinate-frame artifacts, removable by a free linear transform, with the residue bounded by a pre-registered prediction test); with the pairing itself withheld (the correspondence partially identifies itself from shape alone); and constructively (one 512-dimensional linear hub serves up to seven encoders, and a component trained on one transfers to encoders it never saw at 93-99 percent of native performance).

Two things distinguish the project beyond the results. First, its negatives are load-bearing and mechanistic: ICP converging on its own errors, whitening destroying discovery, the hub-width cliff - each measured, each explaining the boundary it marks. Second, its errors are part of the record: five self-corrections reached by measurement rather than argument, which is the difference between a demonstration and research. The final state of the claim is narrower and stronger than the initial one: the shared structure is real, linear, discoverable, and usable - and the coordinate frame that makes it readable is cheap, while the semantic arrangement that makes it meaningful is what training actually buys.

8 · The scale caveat, if it's raised

"Huh et al. go to 70B-parameter language models; your text encoders top out at bge/BERT/GPT-2 scale (768-1024-d). Doesn't that put your alignment numbers in a different, smaller regime?" Yes, and it is worth conceding before it is raised. The effect is already clearly measurable at this scale - the local-structure and objective-over-lineage findings hold on encoders two to three orders of magnitude smaller than the LLMs in the original PRH plots. The closed-form, CPU-only design is the point of this project, not an accident of scale (see the hardware note in Project_Atlas): every result here runs without a GPU past the one-time encoding pass. Scaling the text side is a clean future-work item, and PRH's own capacity-ladder plot predicts it should only strengthen the result, not overturn it. One thing this project does NOT claim: that alignment predicts downstream capability - that correlation is Huh et al.'s result, not measured here. This project's evidence is about the geometry - what is shared, how much, and at what cost to recover it - not about benchmark performance.

One sentence, if needed:

"Huh et al. showed alignment rises with scale and correlates with capability; I took the hypothesis as given and asked what is actually shared, whether one matrix can carry it, and where it fails - and the answers are local structure, yes at 96.7 percent, and at the source encoder's rank."

"Your percentages fall as the encoder gets bigger - isn't that the hub breaking down?" No, and the raw numbers say so. Absolute R@1 rises monotonically: 0.398, 0.442, 0.468. The percentage falls because it is measured against each encoder's OWN native head, and a bigger encoder supports a better one - the denominator rises faster than the numerator. Recovering the natives by division gives 0.357, 0.461, 0.504, and those match E1's separate direct image-to-bge ridge to three decimals on two of three rungs, so the hub costs nothing at the reference point. One real caveat I raise myself: DINOv2-large's entry map is fitted at 4.2 rows per input dimension, below this project's own floor of 5, which biases ridge downward - so 92.9 percent is conservative, not flattering. Recorded in E.13.

"Huh et al. showed alignment rises with scale - wouldn't bigger models converge more?" Tested it. Both modalities, pre-registered. The DINOv2 image ladder (small, base, large, giant) shows rho 0.381, 0.381, 0.418, then 0.339 at giant - rise then fall. The contrastive text ladder (bge-m3 0.568B, Qwen-0.6B, Qwen-4B 4.0B) shows 0.320, 0.352, 0.307 - same shape at 7x the parameters. Both ladders matched pooling, same items, same run. Both pre-registered thresholds falsified. The within-modality picture inverts: contrastive text encoders agree with each other at 0.76-0.86, the highest pairs in the matrix, while their cross-modal agreement sits at 0.19-0.35 and does not improve with scale. Huh et al. worked at a much larger scale (up to 70B) and on language models only; this measures a different regime. Report E.16.

8 · The target-choice objection

"Why bge-m3 as the read-out target when SBERT scores better?" Because SBERT does score better, and the choice is comparability, not merit. The report measures both candidates on the two axes that matter for a read-out target - how well the hub can write into the space, and its own raw retrieval - and SBERT wins on both: writability 101.0 percent against bge's 98.1, raw R@1 0.510 against 0.466 (Table C12f). bge-m3 is retained because every ceiling and every percentage-of-ceiling figure in the project is calibrated against it; switching in the final sections would make those numbers incomparable with Sections 4, C.8, C.10 and C.13, for a gain of about three points and no new insight. SBERT is offered as an alternative in the demo notebook. The report states this in exactly these terms - retained for comparability rather than because the data chose it.

What bge-m3 does contribute, and it is the necessary property: a coordinate frame in which cosine is meaningful. That is what a good read-out target supplies, and it is what BERT (76.6 percent writable, 0.362 raw) and GPT-2 lack. C.11 and C.14 quantify it: whitening a raw language-model space recovers 102.6 percent of the bge arm, and the residual 0.110 spread across the ladder is what contrastive training adds beyond the frame. So the target's advantage is measured, not presupposed - and it is shared by SBERT, which is why the choice between the two comes down to comparability.

Land on: "SBERT is marginally better on both measured axes. bge-m3 is kept because every figure in the report is calibrated against it, and the cost of that choice - about three points - is disclosed."

Trap: Do not claim bge-m3 won on merit. It did not, the report says so, and an examiner who has read C.13.4 will catch it. Volunteering that SBERT is better is the stronger move - it shows the inconvenient measurement was made and reported.

"So SBERT is cleaner geometrically too - isn't that a third strike against bge?" It is a third measured difference - bge's mean pair-cosine is +0.57 against SBERT's +0.135, so bge carries a shared background direction that SBERT largely lacks. But it is not a third independent cost. The anisotropy is the CAUSE of the two gaps already on the table: bge writes at 98.1 rather than 101 percent, and retrieves at 0.466 rather than 0.510, BECAUSE of that background direction. Counting the anisotropy separately double-counts one fact. And bge clears the only bar that disqualifies a target - a map ranks items inside it at 98.1 percent, nowhere near GPT-2's 24.8 percent failure; anisotropic is not collapsed, and its effective rank of 97 confirms the structure is intact beneath the shared axis.

The reversal, and it is the point to land: bge being the harder, anisotropic target makes every result MORE conservative, not weaker. Each figure in the project is a percentage of a bge ceiling; a cleaner SBERT ceiling would sit higher and pull every reported share - 94.2, 96.7 percent and the rest - DOWN. Measuring against the slightly tougher target means the numbers are if anything understated.